

Spacecraft Attitude Dynamics & Control

우주비행체의 자세 동역학 및 제어

16th Short Course on Space Technology

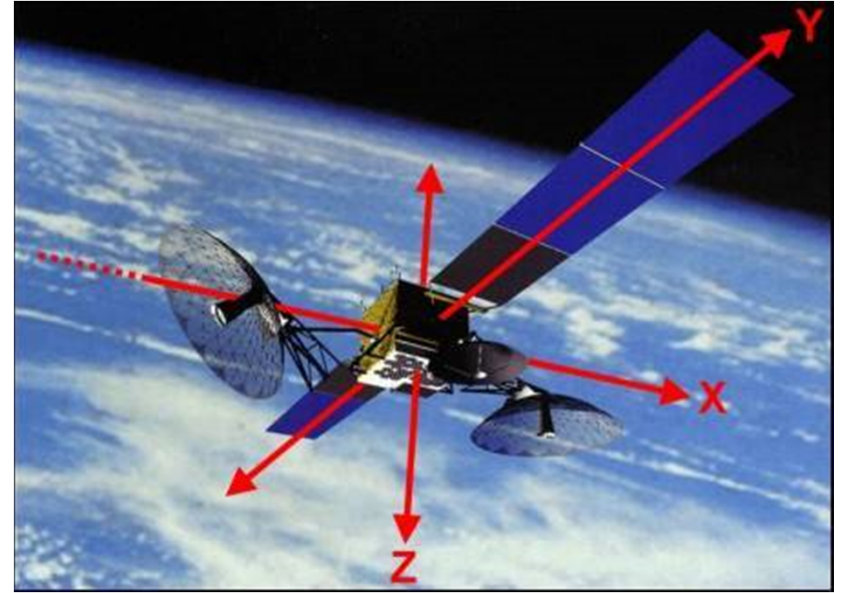
June 30 – July 1, 2011

Chandeok Park
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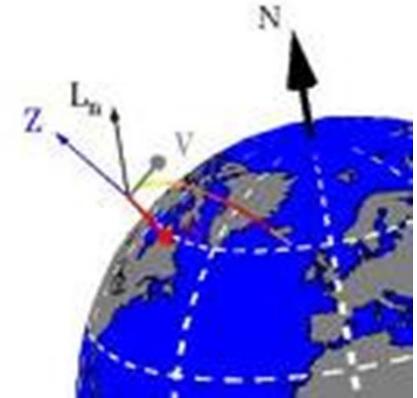
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Introduction

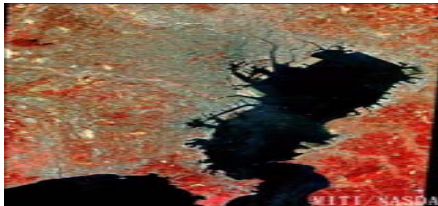


Satellite Systems: Bus & Payload

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관측위성

전자광학 카메라



통신방송위성

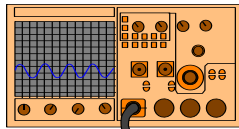
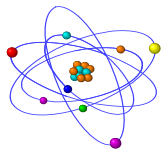
트랜스폰더



탑재체

과학위성

과학 탑재체



과학
실험

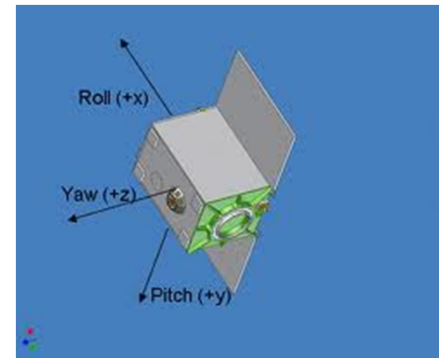
위성 본체

- 구조 및 열제어계
- 전력계 (태양 전지판 포함)
- 자세 제어계
- 추진계
- 원격측정 및 명령계

Attitude ? Attitude Control? What for?

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- Attitude = *Orientation of Spacecraft*
- Attitude Control is to keep/maneuver the spacecraft to a desired orientation
 - Stabilization/Reorientation
- What is 'Attitude Control' for?
 - Remote Sensing: to Point a camera to the target at Earth Surface
 - Communication: to Point an antenna to the ground station
 - Space Science: to Point a science instrument to the celestial body



Pointing Accuracy Requirements

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- Modern Spacecraft Missions require more accurate and agile attitude control

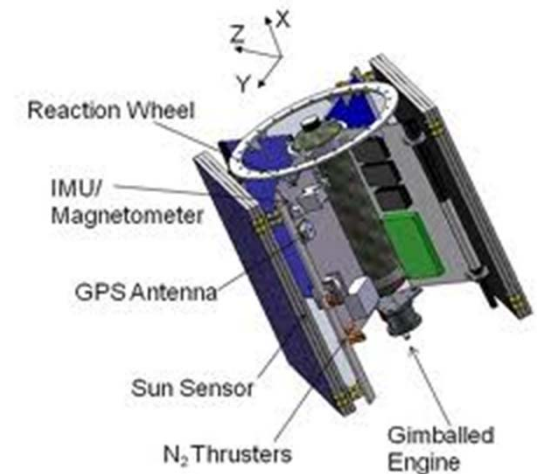
Mission	Payload	Pointing Accuracy (deg)	Pointing Stability (deg/sec)
Communications	Transponder / Antenna	0.5 ~ 0.05	0.1 ~ 0.05
Earth Observation Surveillance	Imaging Device	0.1 ~ 0.01	0.5 ~ 0.005
Interplanetary	Imaging / Scientific Instruments	0.5 ~ 0.05	0.05 ~ 0.005
Scientific	Scientific Instruments	0.5 ~ 0.01	0.1 ~ 0.05
Astronomical	Imaging Devices	0.5 ~ 10 (arcsec)	0.5 ~ 0.05 (arcsec)

Attitude Control Systems Overview

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- Three Basic Elements

- Sensors: Measure attitude error
- Control Process: Compute corrective action
- Actuators: Execute corrections

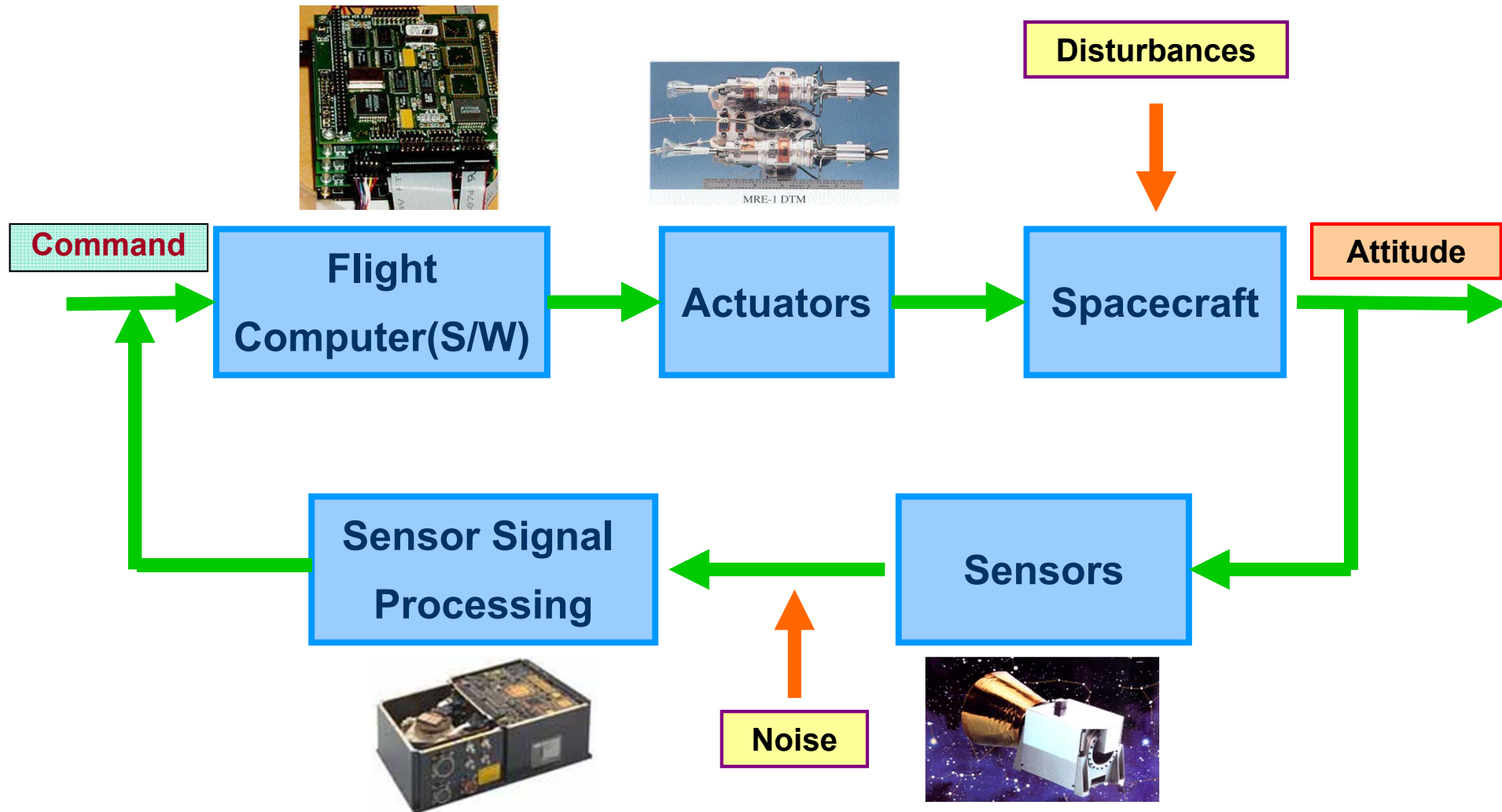


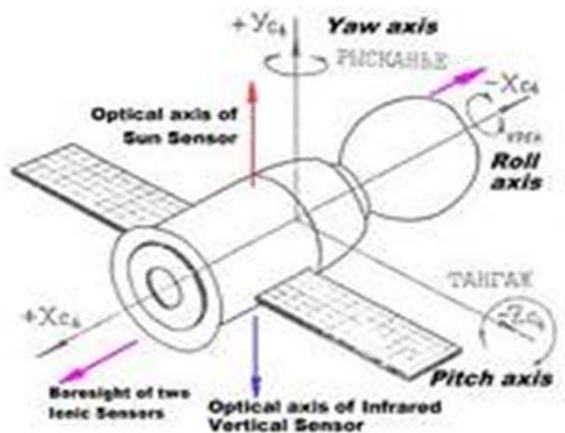
- Attitude control systems are highly mission dependent

- Mission pointing requirements
- Mission orbital characteristics
- Control system stability and response time
- Interaction of control systems with onboard equipments
- Power requirements
- Weight restrictions

Attitude Control Systems Overview

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Attitude Representations



Attitude = Orientation of S/C

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Attitude Representations	No. of Parameters	Characteristics
Direction Cosine Matrix =Rotation Matrix	9	Nonsingular
Euler Principal Angle & Axis	4	Nonsingular
Euler Angles	3	Singular (Gimbal Lock) Physical Interpretation
Quaternion	4	Nonsingular Computational Convenience
Rodrigues Parameters (Gibbs Vector)	3	Singular (180° rotation) Useful for Analytical Study
Modified Rodrigues Parameters	3	Singular (360° rotation) Useful for Analytic Study

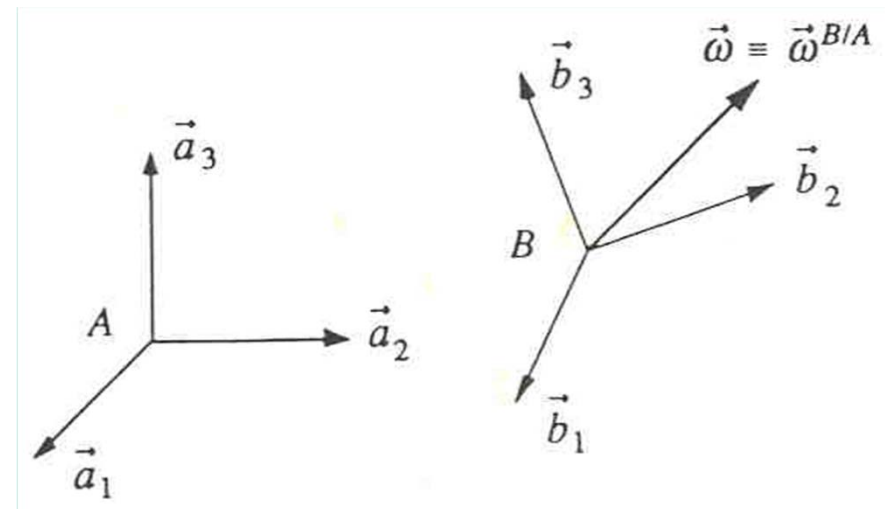
- Linked to each other by Mathematical Relations

AR 1: Direction Cosine (Rotation) Matrix

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- Projection of rotating unit vectors of Frame B onto the fixed unit vectors of Frame A
- Specifying the coordinates of rotating unit vectors of Frame B in terms of the fixed Frame A
- C is orthogonal, i.e., $R^{-1}=R^T$

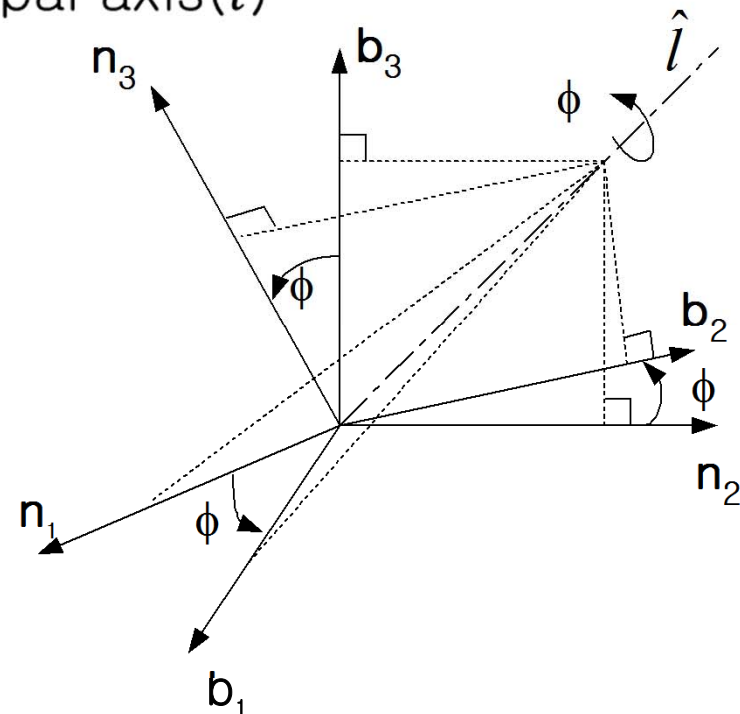
$$C = \begin{bmatrix} \vec{b}_1 \cdot \vec{a}_1 & \vec{b}_1 \cdot \vec{a}_2 & \vec{b}_1 \cdot \vec{a}_3 \\ \vec{b}_2 \cdot \vec{a}_1 & \vec{b}_2 \cdot \vec{a}_2 & \vec{b}_2 \cdot \vec{a}_3 \\ \vec{b}_3 \cdot \vec{a}_1 & \vec{b}_3 \cdot \vec{a}_2 & \vec{b}_3 \cdot \vec{a}_3 \end{bmatrix}$$



AR 2: Euler Principal Angle & Axis

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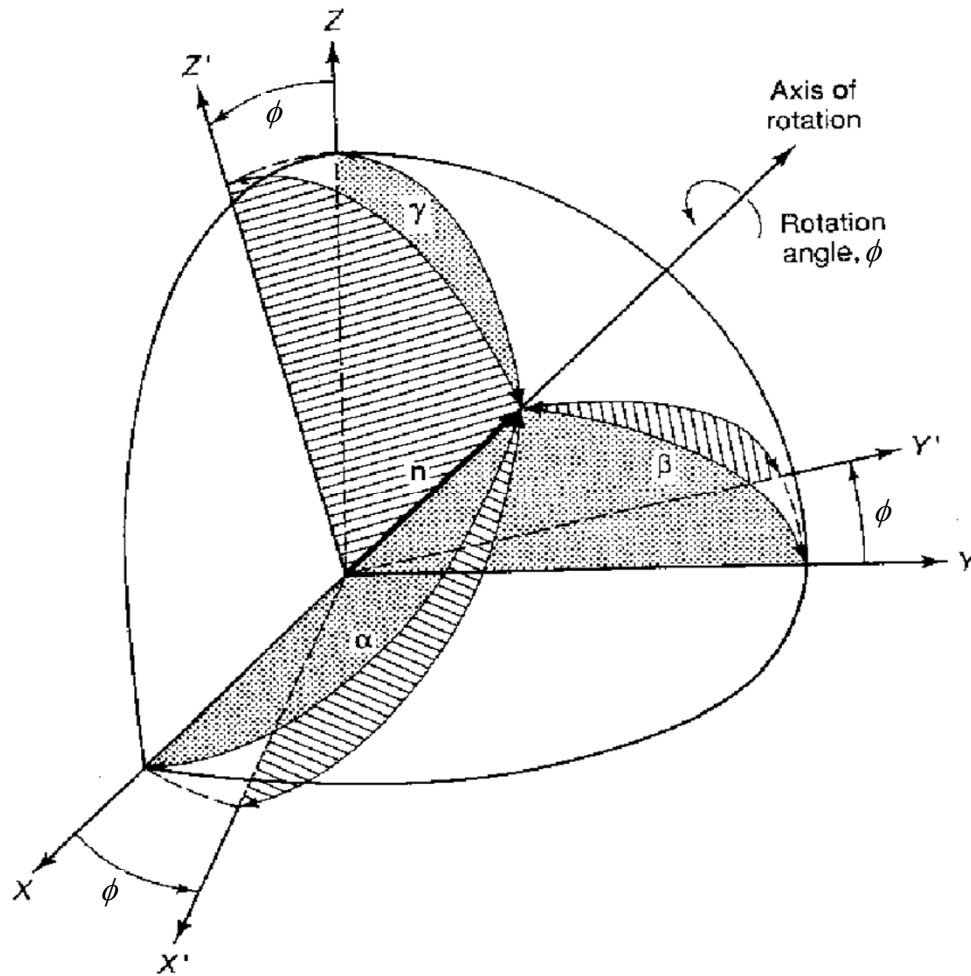
- Euler's Principal Rotation Theorem
 - A rigid body can be brought from an arbitrary initial orientation to an arbitrary orientation by a single rotation through a principal angle(ϕ) about a principal axis($\hat{l}$)
 - $\hat{l}$ is unity by convention
 - Rotation Vector: $\vec{v} = \phi \hat{l}$



AR 3: Quaternion

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- ϕ : principal angle



$$q_1 = \sin \frac{\phi}{2} \cos \alpha$$

$$q_2 = \sin \frac{\phi}{2} \cos \beta$$

$$q_3 = \sin \frac{\phi}{2} \cos \gamma$$

$$q_4 = \cos \frac{\phi}{2}$$

$$\mathbf{q} = q_4 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}$$

$$= q_4 + \mathbf{q}$$

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1$$

AR 4: Euler Angles

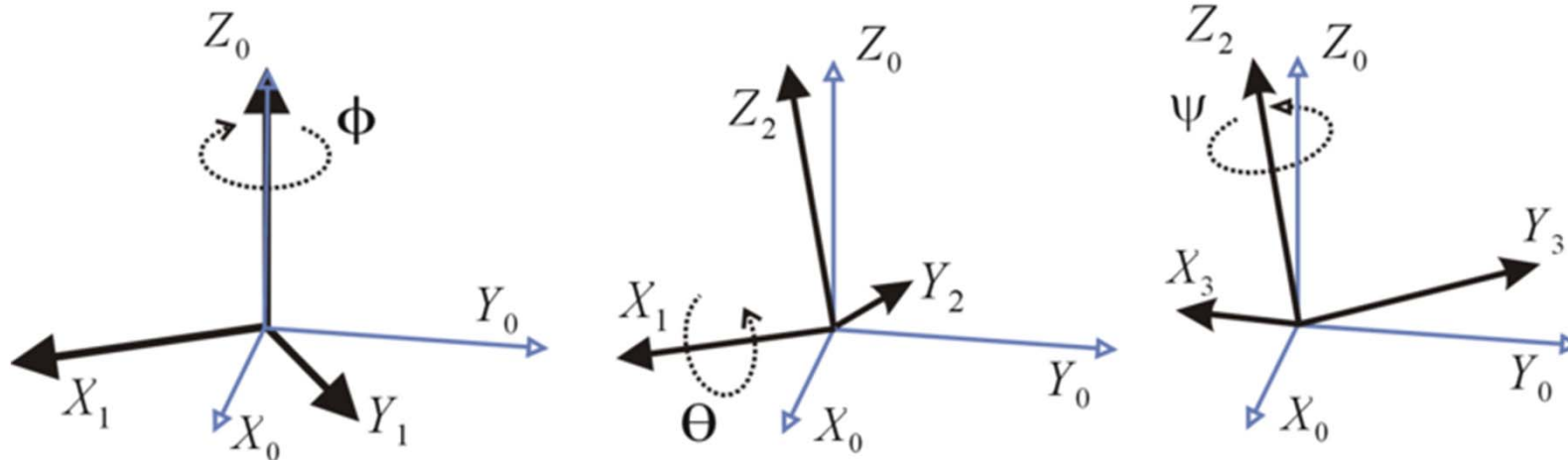
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- 3 sequential rotations about 3 axes (ψ , θ , ϕ)
 - 12 combinations
 - Gimbal Lock singularity

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



AR 5-6: Rodrigues Parameters, etc

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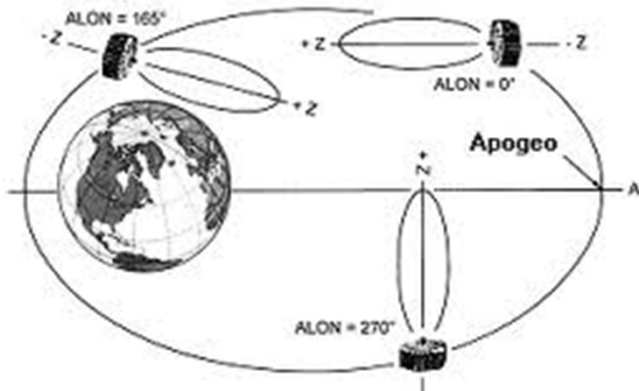
- Rodrigues Parameters = Gibbs Vector

- $\vec{g} = \hat{l} \tan \phi / 2 = \vec{q} / q_4$
- Useful for analytical study
- Undefined for 180° rotations
- Undesirable for global attitude representation

- Modified Rodrigues Parameters

- $\vec{p} = \hat{l} \tan \phi / 4 = \vec{q} / (1 + q_4)$
- Useful for analytical study
- Undefined for 360° rotations





Kinematics

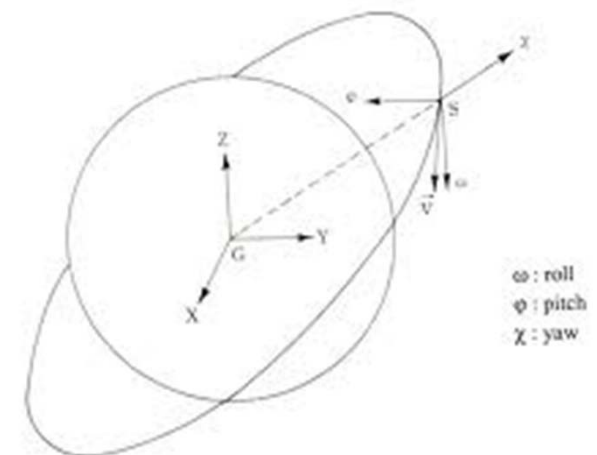


Figure 5.3.2 Satellite attitude parameters

Rotation of Reference Frame

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- $\vec{\omega}^{B/A}$: Angular Velocity of Frame B w.r.t. the Frame A
- Time derivatives of the unit vectors $\vec{b}_i$ w.r.t. Frame A

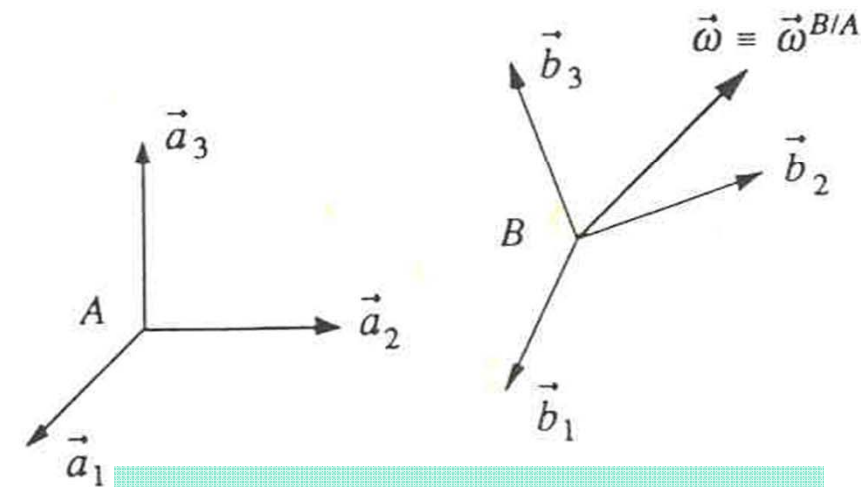
$$\frac{d\vec{b}_i}{dt} = \vec{\omega}^{B/A} \times \vec{b}_i$$

- Perpendicular to the plane generated by $\vec{\omega}^{B/A}$ and $\vec{b}_i$

- Let $\vec{\omega} = \vec{\omega}^{B/A} = \omega_1 \vec{b}_1 + \omega_2 \vec{b}_2 + \omega_3 \vec{b}_3$

Then,

$$\begin{pmatrix} \frac{d\vec{b}_1}{dt} \\ \frac{d\vec{b}_2}{dt} \\ \frac{d\vec{b}_3}{dt} \end{pmatrix} = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix} \begin{pmatrix} \vec{b}_1 \\ \vec{b}_2 \\ \vec{b}_3 \end{pmatrix}$$

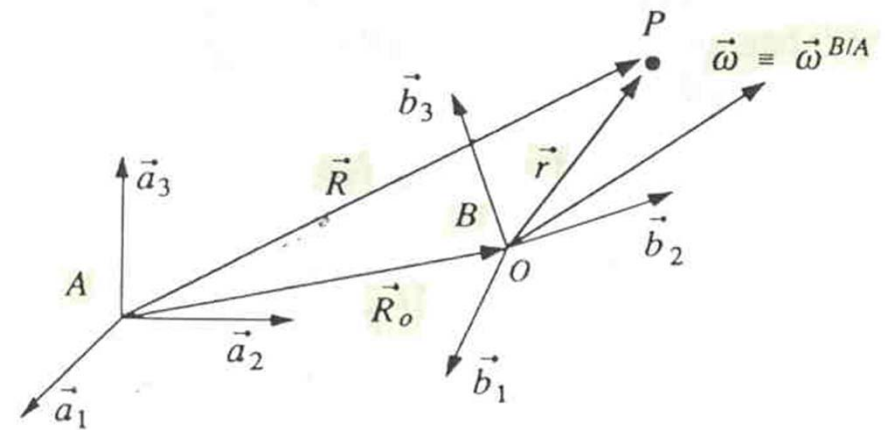


Reference Frames A and B

Kinematics in Rotating Frame

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$$\begin{aligned}
 \vec{r} &= r_1 \vec{b}_1 + r_2 \vec{b}_2 + r_3 \vec{b}_3 \\
 \left(\frac{d\vec{r}}{dt} \right)_A &= \left(\dot{r}_1 \vec{b}_1 + \dot{r}_2 \vec{b}_2 + \dot{r}_3 \vec{b}_3 \right) + \left(r_1 \dot{\vec{b}}_1 + r_2 \dot{\vec{b}}_2 + r_3 \dot{\vec{b}}_3 \right) \\
 &= \left(\frac{d\vec{r}}{dt} \right)_B + \left[r_1 (\vec{\omega} \times \vec{b}_1) + r_2 (\vec{\omega} \times \vec{b}_2) + r_3 (\vec{\omega} \times \vec{b}_3) \right] \\
 &= \left(\frac{d\vec{r}}{dt} \right)_B + \vec{\omega} \times \vec{r}
 \end{aligned}$$



Reference Frames A and B

$$\begin{aligned}
 \vec{R} &= \vec{R}_o + \vec{r} \\
 \left(\frac{d\vec{R}}{dt} \right)_A &= \left(\frac{d\vec{R}_o}{dt} \right)_A + \left(\frac{d\vec{r}}{dt} \right)_A = \left(\frac{d\vec{R}_o}{dt} \right)_A + \left(\frac{d\vec{r}}{dt} \right)_B + \vec{\omega} \times \vec{r} \\
 \left(\frac{d^2\vec{R}}{dt^2} \right)_A &= \left(\frac{d^2\vec{R}_o}{dt^2} \right)_A + \left(\frac{d^2\vec{r}}{dt^2} \right)_B + \dot{\vec{\omega}} \times \vec{r} + 2\vec{\omega} \times \left(\frac{d\vec{r}}{dt} \right)_B + \vec{\omega} \times (\vec{\omega} \times \vec{r})
 \end{aligned}$$

Attitude Kinematics 1: DCM

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- Let the skew-symmetric Angular Velocity Tensor be

$$\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} = -\Omega^T$$

- $\{\mathbf{b}\} = \mathbf{C}\{\mathbf{a}\}$

$$\begin{aligned} \{\dot{\mathbf{b}}\} &= \dot{\mathbf{C}}\{\mathbf{a}\} \Leftrightarrow \{\dot{\mathbf{b}}\} = \Omega\{\mathbf{b}\} = \Omega\mathbf{C}\{\mathbf{a}\} \\ \Rightarrow \dot{\mathbf{C}} &= \Omega\mathbf{C} \end{aligned}$$

Attitude Kinematics 2: Quaternion

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- Start from DCM kinematics:

$$\dot{C} = \Omega C$$

- Introduce DCM $\leftrightarrow$ Quaternion relation, $f(q, \dot{q})$ $f(\omega)$ $f(q)$

$$\omega_1 = 2(\dot{q}_1 q_4 + \dot{q}_2 q_3 - \dot{q}_3 q_2 - \dot{q}_4 q_1)$$

$$\omega_2 = 2(\dot{q}_2 q_4 + \dot{q}_3 q_1 - \dot{q}_1 q_3 - \dot{q}_4 q_2)$$

$$\omega_3 = 2(\dot{q}_3 q_4 + \dot{q}_1 q_2 - \dot{q}_2 q_1 - \dot{q}_4 q_3)$$

- Differentiate the identity $q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1$

$$2(\dot{q}_1 q_1 + \dot{q}_2 q_2 + \dot{q}_3 q_3 + \dot{q}_4 q_4) = 0$$

Attitude Kinematics 2: Quaternion

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- Group 4 scalar differential equations

$$\begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix} = 2 \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \\ q_1 & q_2 & q_3 & q_4 \end{bmatrix} \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{Bmatrix} \Leftrightarrow \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & -q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix}$$

$$\Leftrightarrow \begin{Bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{Bmatrix}$$

$$\Leftrightarrow \dot{\mathbf{q}} = \frac{1}{2} \boldsymbol{\omega} \times \mathbf{q} + \frac{1}{2} q_4 \boldsymbol{\omega}, \quad \dot{q}_4 = -\frac{1}{2} \boldsymbol{\omega}^T \mathbf{q}$$

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}, \quad \boldsymbol{\omega} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

Attitude Kinematics 3: Euler Angles

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- Example: 3–2–1 rotation (out of 12 combinations)

- Start from the inertial frame
- Rotate sequentially about the intermediate axis

$$\begin{aligned}
 \begin{Bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{Bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_3 & \sin \theta_3 \\ 0 & -\sin \theta_3 & \cos \theta_3 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & 0 & -\sin \theta_2 \\ 0 & 1 & 0 \\ \sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix} \begin{bmatrix} \cos \theta_1 & \sin \theta_1 & 0 \\ -\sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{Bmatrix} \\
 &= \begin{bmatrix} \cos \theta_2 \cos \theta_1 & \cos \theta_2 \sin \theta_1 & -\sin \theta_2 \\ \sin \theta_3 \sin \theta_2 \cos \theta_1 - \cos \theta_3 \sin \theta_1 & \sin \theta_3 \sin \theta_2 \sin \theta_1 + \cos \theta_3 \cos \theta_1 & \sin \theta_3 \cos \theta_2 \\ \cos \theta_3 \sin \theta_2 \cos \theta_1 + \sin \theta_3 \sin \theta_1 & \cos \theta_3 \sin \theta_2 \sin \theta_1 - \sin \theta_3 \cos \theta_1 & \cos \theta_3 \cos \theta_2 \end{bmatrix} \begin{Bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{Bmatrix} \\
 &= [C(\theta_1, \theta_2, \theta_3)] \begin{Bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{Bmatrix}
 \end{aligned}$$

$$[C(\theta_1, \theta_2, \theta_3)] = [C_i(\theta_3)][C_j(\theta_2)][C_k(\theta_1)]$$

Attitude Kinematics 3: Euler Angles

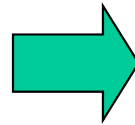
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- 3 successive rotations of Euler rates

$$C_3(\theta_3) : A' \leftarrow A$$

$$C_2(\theta_2) : A'' \leftarrow A'$$

$$C_1(\theta_1) : B \leftarrow A''$$

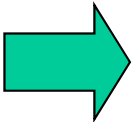


$$\vec{\omega}^{A'/A} = \dot{\theta}_3 \vec{a}_3 = \dot{\theta}_3 \vec{a}'_3$$

$$\vec{\omega}^{A''/A'} = \dot{\theta}_2 \vec{a}'_2 = \dot{\theta}_2 \vec{a}''_2$$

$$\vec{\omega}^{B/A''} = \dot{\theta}_1 \vec{a}''_1 = \dot{\theta}_1 \vec{b}_1$$

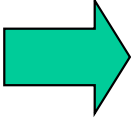
$$\vec{\omega}^{B/A} = \vec{\omega}^{B/A''} + \vec{\omega}^{A''/A'} + \vec{\omega}^{A'/A} = \dot{\theta}_1 \vec{b}_1 + \dot{\theta}_2 \vec{a}''_2 + \dot{\theta}_3 \vec{a}'_3$$

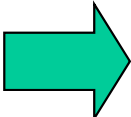


$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} \dot{\theta}_1 \\ 0 \\ 0 \end{bmatrix} + C_1(\theta_1) \begin{bmatrix} 0 \\ \dot{\theta}_2 \\ 0 \end{bmatrix} + C_1(\theta_1)C_2(\theta_2) \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix}$$

Attitude Kinematics 3: Euler Angles

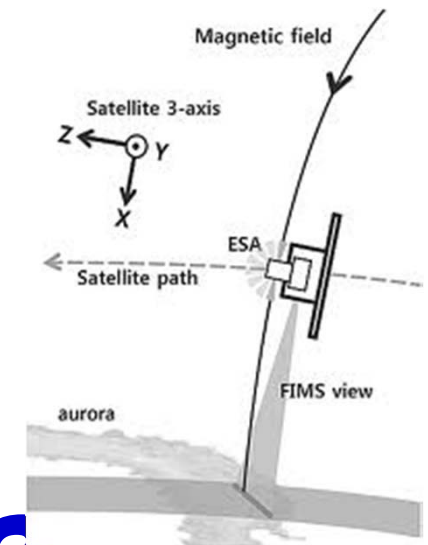
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$$\begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta_2 \\ 0 & \cos \theta_1 & \sin \theta_1 \cos \theta_2 \\ 0 & -\sin \theta_1 & \cos \theta_1 \cos \theta_2 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{Bmatrix}$$


$$\begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{Bmatrix} = \frac{1}{\cos \theta_2} \begin{bmatrix} \cos \theta_2 & \sin \theta_1 \sin \theta_2 & \cos \theta_1 \sin \theta_2 \\ 0 & \cos \theta_1 \cos \theta_2 & -\sin \theta_1 \cos \theta_2 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix}$$

- Singularity at $\theta_2 = \pm \pi/2$

Dynamics & Stability Analysis

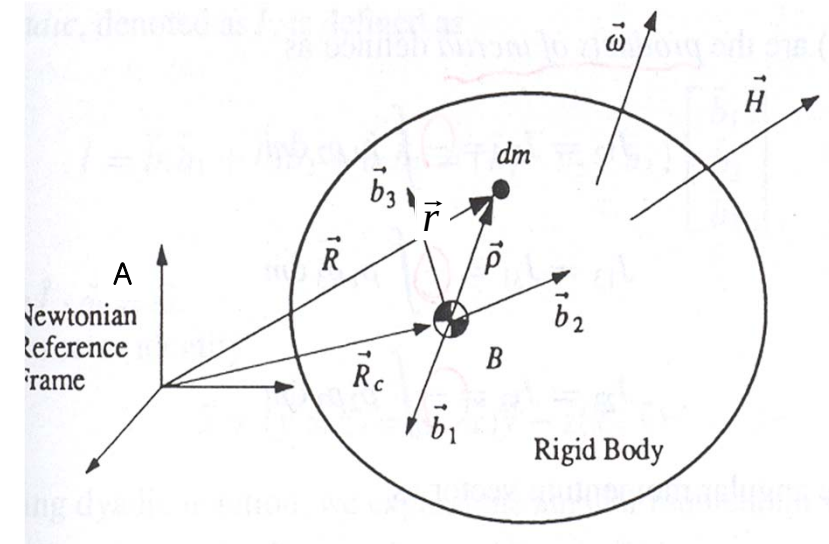


Newtonian Law of Motion

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- Angular Momentum

$$\vec{H} = \int [\vec{r} \times (\vec{\omega} \times \vec{r})] dm$$



- Newton's 2nd Law

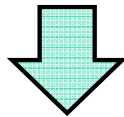
- (Time rate of change of angular momentum)
= (Sum of all external moments acted on the rigid body)

$$\left(\frac{d\vec{H}}{dt} \right)_A = \left(\frac{d\vec{H}}{dt} \right)_B + \vec{\omega} \times \vec{H} = \sum_i M_i$$

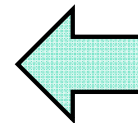
Angular Momentum for Rigid Body

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$$\vec{H} = \int [\vec{r} \times (\vec{\omega} \times \vec{r})] dm$$



$$\begin{aligned} \vec{H} = & \int_M \left\{ \omega_1 [r_2^2 + r_3^2] - \omega_2 r_1 r_2 - \omega_3 r_1 r_3 \right\} dm \vec{b}_1 \\ & + \int_M \left\{ \omega_2 [r_3^2 + r_1^2] - \omega_3 r_2 r_3 - \omega_1 r_1 r_2 \right\} dm \vec{b}_2 \\ & + \int_M \left\{ \omega_3 [r_1^2 + r_2^2] - \omega_1 r_1 r_3 - \omega_2 r_2 r_3 \right\} dm \vec{b}_3 \end{aligned}$$



$$\begin{aligned} J_1 &= \int_M (r_2^2 + r_3^2) dm & J_1 + J_2 &> J_3 \\ J_2 &= \int_M (r_3^2 + r_1^2) dm & J_2 + J_3 &> J_1 \\ J_3 &= \int_M (r_1^2 + r_2^2) dm & J_3 + J_1 &> J_2 \\ J_{ij} &= - \int_M r_i r_j dm & (i, j = 1, 2, 3, i \neq j) \end{aligned}$$

Moment of Inertia

Product of Inertia

$$\vec{H} = \begin{bmatrix} J_1 & J_{12} & J_{13} \\ J_{21} & J_2 & J_{23} \\ J_{31} & J_{32} & J_3 \end{bmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \equiv \mathcal{J} \vec{\omega}$$

Rigid Body Equations of Motion

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$$\left(\frac{d\vec{H}}{dt} \right)_A = \left(\frac{d\vec{H}}{dt} \right)_B + \vec{\omega} \times \vec{H} = J \dot{\vec{\omega}} + \vec{\omega} \times \vec{H} = \sum_i M_i$$

● General Equations of Motion

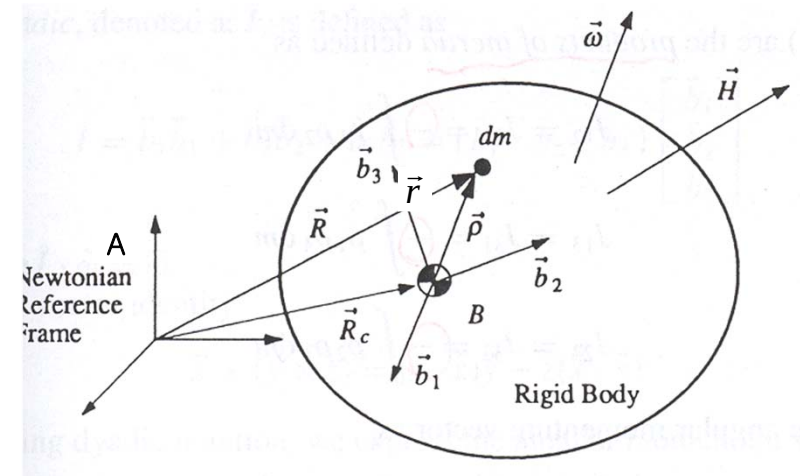
$$\begin{bmatrix} J_1 & J_{12} & J_{13} \\ J_{21} & J_2 & J_{23} \\ J_{31} & J_{32} & J_3 \end{bmatrix} \begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{pmatrix} + \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} J_1 & J_{12} & J_{13} \\ J_{21} & J_2 & J_{23} \\ J_{31} & J_{32} & J_3 \end{bmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} M_1 \\ M_2 \\ M_3 \end{pmatrix}$$

● In case of Principal Axis ($J_{ij} = 0$)

$$J_1 \dot{\omega}_1 - (J_2 - J_3) \omega_2 \omega_3 = M_1$$

$$J_2 \dot{\omega}_2 - (J_3 - J_1) \omega_3 \omega_1 = M_2$$

$$J_3 \dot{\omega}_3 - (J_1 - J_2) \omega_1 \omega_2 = M_3$$



Torque-free Motion (General)

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$$J_1 \dot{\omega}_1 - (J_2 - J_3) \omega_2 \omega_3 = 0$$

$$J_2 \dot{\omega}_2 - (J_3 - J_1) \omega_3 \omega_1 = 0$$

$$J_3 \dot{\omega}_3 - (J_1 - J_2) \omega_1 \omega_2 = 0$$



- Conservation of Angular Momentum

$$H^2 = J_1^2 \omega_1^2 + J_2^2 \omega_2^2 + J_3^2 \omega_3^2$$

- Conservation of Energy

$$2E = J_1 \omega_1^2 + J_2 \omega_2^2 + J_3 \omega_3^2$$



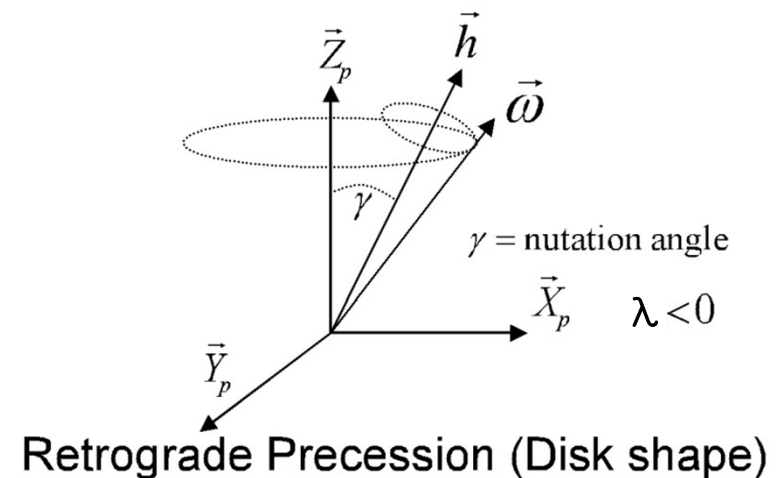
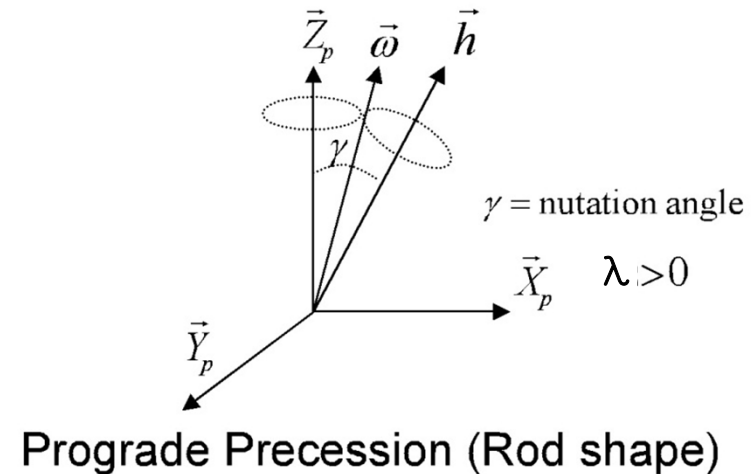
Torque-free Motion (Axisymmetric; $J_1=J_2$)

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$$\begin{aligned} J_1 \dot{\omega}_1 - (J_1 - J_3) \omega_2 \omega_3 &= 0 \\ J_1 \dot{\omega}_2 - (J_3 - J_1) \omega_3 \omega_1 &= 0 \\ J_3 \dot{\omega}_3 &= 0 \end{aligned}$$

$$\begin{aligned} \omega_x(t) &= \omega_{x0} \cos \lambda t + \omega_{y0} \sin \lambda t \\ \omega_y(t) &= \omega_{y0} \cos \lambda t - \omega_{x0} \sin \lambda t \\ \omega_z(t) &= \omega_{z0}, \quad \lambda = (J_1 - J_3) \omega_{z0} / J_1 \end{aligned}$$

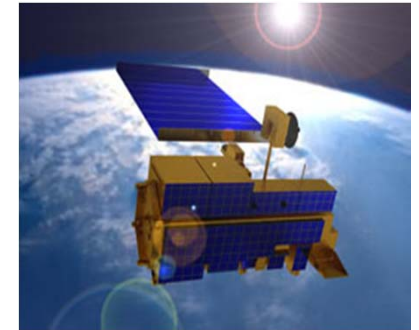
- Prograde Precession
 - If $J_1=J_2 > J_3$ (rod-shaped)
- Retrograde Precession
 - If $J_1=J_2 < J_3$ (disc-shaped)
- body cone vs. space cone



Rigid S/C in a Circular Orbit

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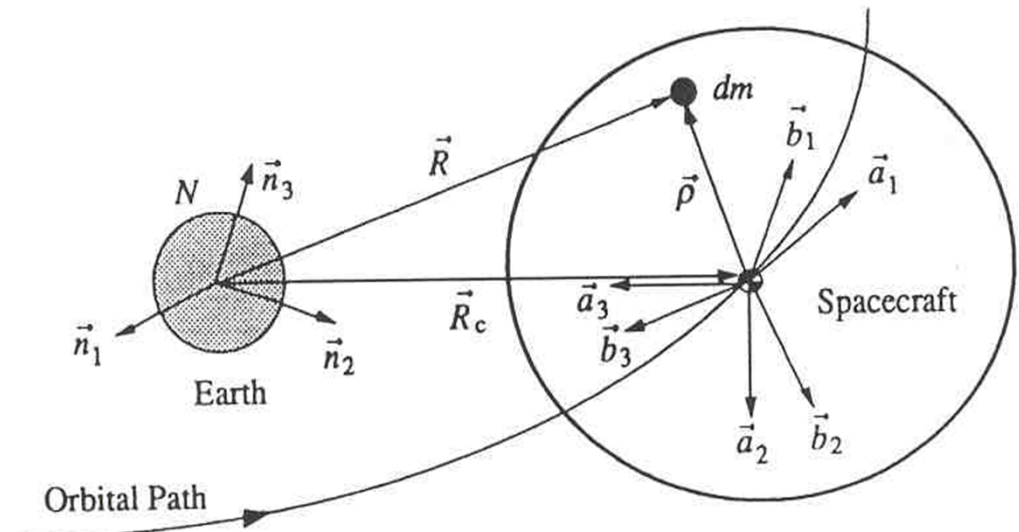
- Effect of Gravitational Forces/Moments
 - First considered by D'Alembert and Euler



- Gravitational Forces
/Gravity-gradient Torque

$$d\vec{f} = -\frac{\mu \vec{R} dm}{|\vec{R}|^3} = -\frac{\mu (\vec{R}_c + \vec{\rho}) dm}{|\vec{R}_c + \vec{\rho}|^3}$$

$$\vec{M} = \int \vec{\rho} \times d\vec{f} = -\mu \int \frac{(\vec{R}_c + \vec{\rho})}{|\vec{R}_c + \vec{\rho}|^3} dm$$



Newtonian frame N
LVLH frame A
Body-fixed frame B

$$\vec{\omega}^{A/N} = -n \vec{a}_2$$

$$\vec{b} = C^{B/A} \vec{a}$$

Rigid S/C in a Circular Orbit: EOM

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- Kinematic Equations of Motion (3–2–1 rotation)

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} = \frac{1}{\cos \theta_2} \begin{bmatrix} \cos \theta_2 & \sin \theta_1 \sin \theta_2 & \cos \theta_1 \sin \theta_2 \\ 0 & \cos \theta_1 \cos \theta_2 & \sin \theta_1 \cos \theta_2 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix} + \frac{n}{\cos \theta_2} \begin{bmatrix} \sin \theta_3 \\ \cos \theta_2 \cos \theta_3 \\ \sin \theta_2 \sin \theta_3 \end{bmatrix}$$

- Equations of Motion for Body-fixed Principal Axis

$$J_1 \dot{\omega}_1 - (J_2 - J_3) \omega_2 \omega_3 = -3n^2 (J_2 - J_3) C_{23} C_{33}$$

$$J_2 \dot{\omega}_2 - (J_3 - J_1) \omega_3 \omega_1 = -3n^2 (J_3 - J_1) C_{33} C_{13}$$

$$J_3 \dot{\omega}_3 - (J_1 - J_2) \omega_1 \omega_2 = -3n^2 (J_1 - J_2) C_{13} C_{23}$$

Gravity-gradient Torque

- Linearized Equations of Motion

$$J_1 \ddot{\theta}_1 - n(J_1 - J_2 + J_3) \dot{\theta}_3 + 4n^2 (J_2 - J_3) \theta_1 = 0$$

$$J_2 \ddot{\theta}_2 + 3n^2 (J_1 - J_3) \theta_2 = 0$$

$$J_3 \ddot{\theta}_3 + n(J_1 - J_2 + J_3) \dot{\theta}_1 + n^2 (J_2 - J_1) \theta_3 = 0$$

Decoupled Pitch Motion

Rigid S/C in a Circular Orbit: Stability

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● Pitch Axis Stability

- Characteristic Equation : $s^2 + [3n^2(J_1 - J_3)/J_2] = 0$
- $J_1 < J_3$: one unstable pole
- $J_1 > J_3$: 2 poles on imaginary axis → *neutrally stable*

● Roll/Yaw Axis Stability

- Characteristic Equation : $s^4 + (1 + 3k_1 + k_1k_3)n^2s^2 + 4k_1k_3n^4 = 0$
- Roll/Yaw Axis Stability Criteria : 4 poles on imaginary axis

$$\begin{aligned} k_1k_3 &> 0 \\ 1 + 3k_1 + k_1k_3 &> 0 \\ (1 + 3k_1 + k_1k_3)^2 - 16k_1k_3 &> 0 \end{aligned}$$

$$k_1 = (J_2 - J_3)/J_1$$

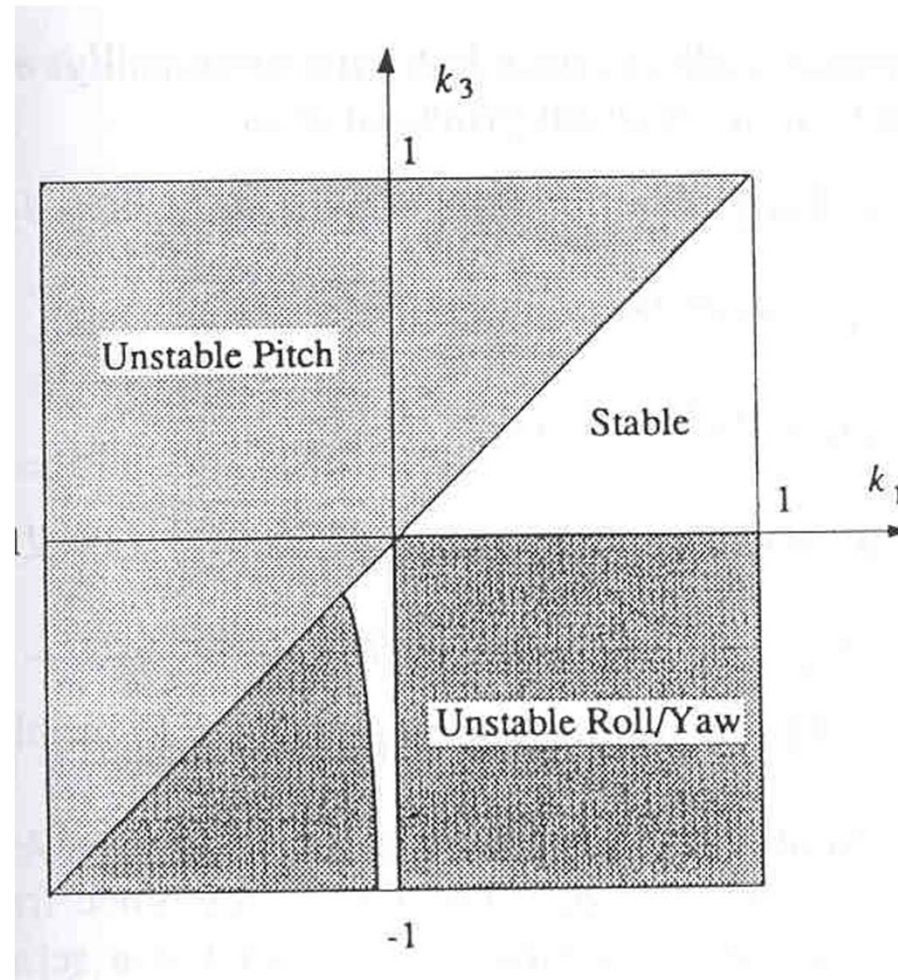
$$k_3 = (J_2 - J_1)/J_3$$

$$|k_1| < 1, |k_3| < 1$$

Rigid S/C in a Circular Orbit: Stability

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- Gravity–gradient Stability Plot



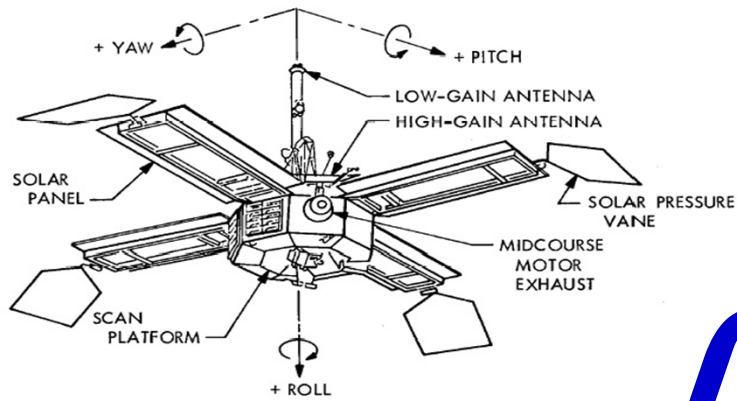


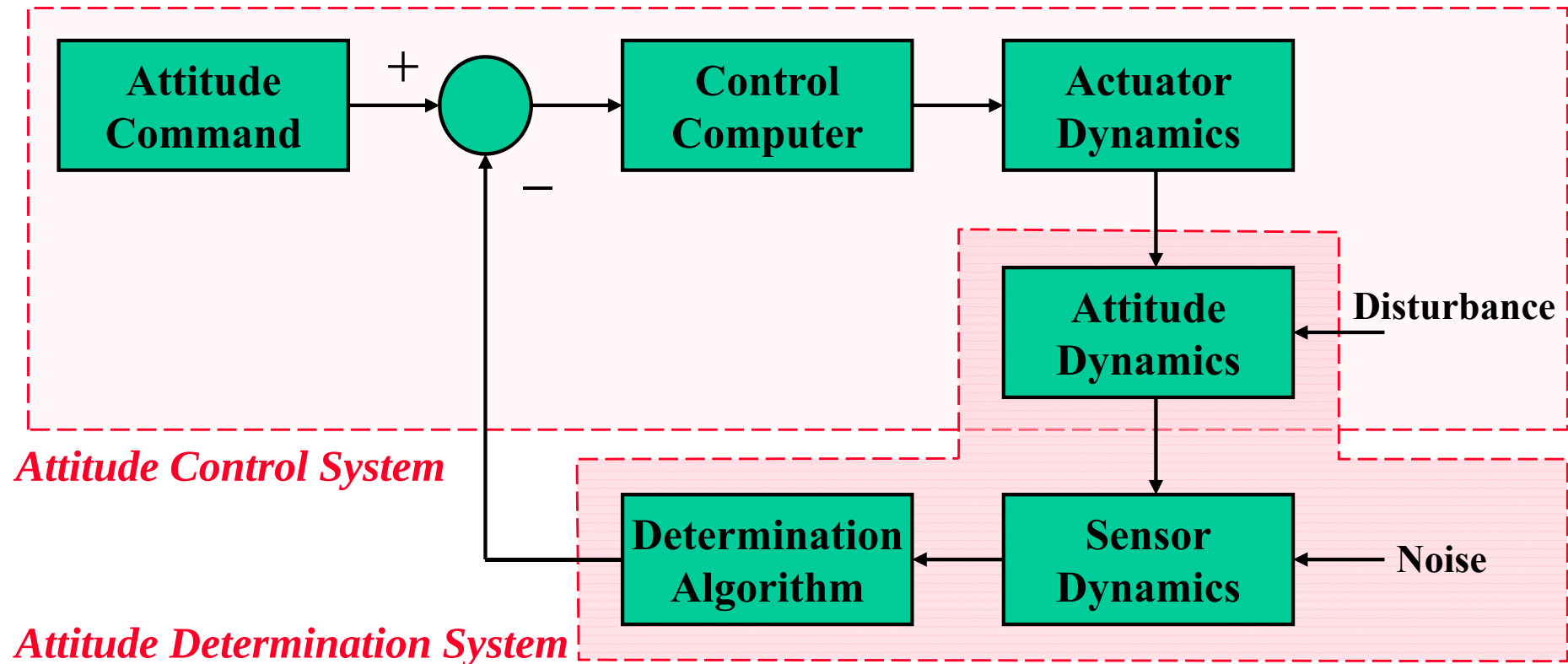
Figure 12.—Mariner 4.

Control : Stabilization & Reorientation



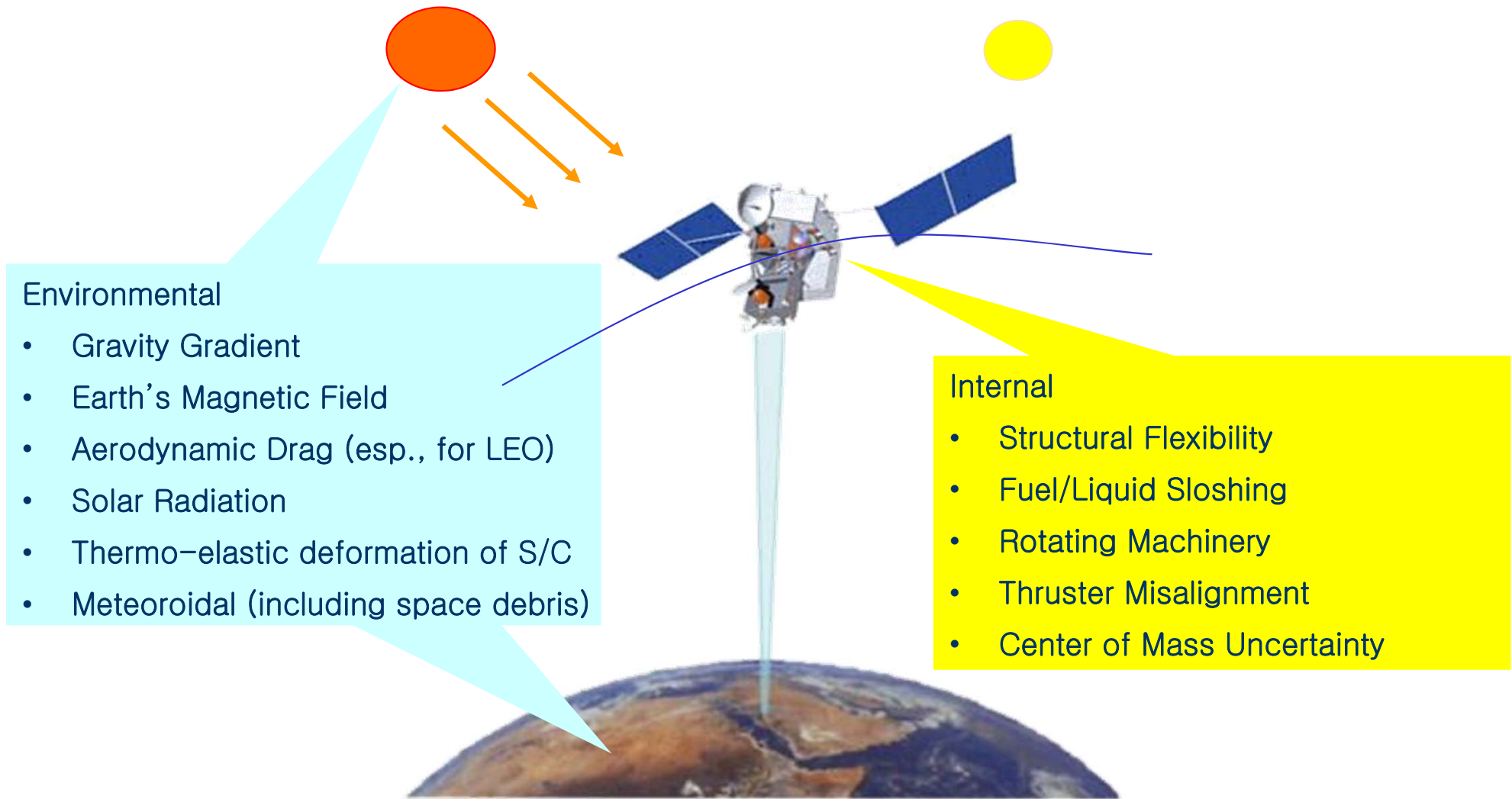
Attitude Determination & Control System

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Disturbances

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Stabilization Techniques

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● Passive

- Gravity Gradient
- Magnetic Torque
- Nutation Damping
 - Driven by the motion itself
 - Dissipates energy
- Fuel Sloshing

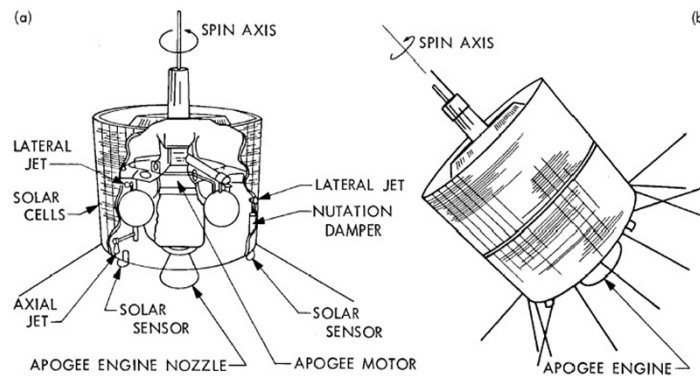
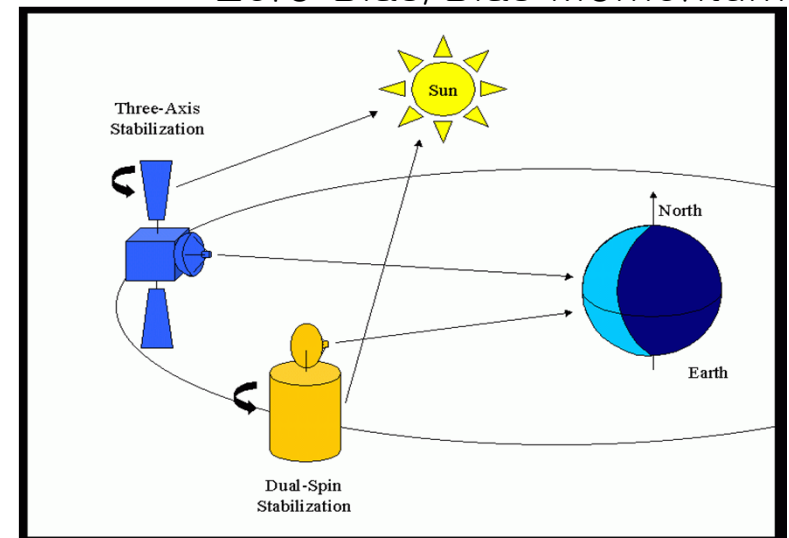


Figure 1.—Single-body spin-stabilized spacecraft: (a) Syncom spacecraft; (b) ATS spacecraft.

● Active

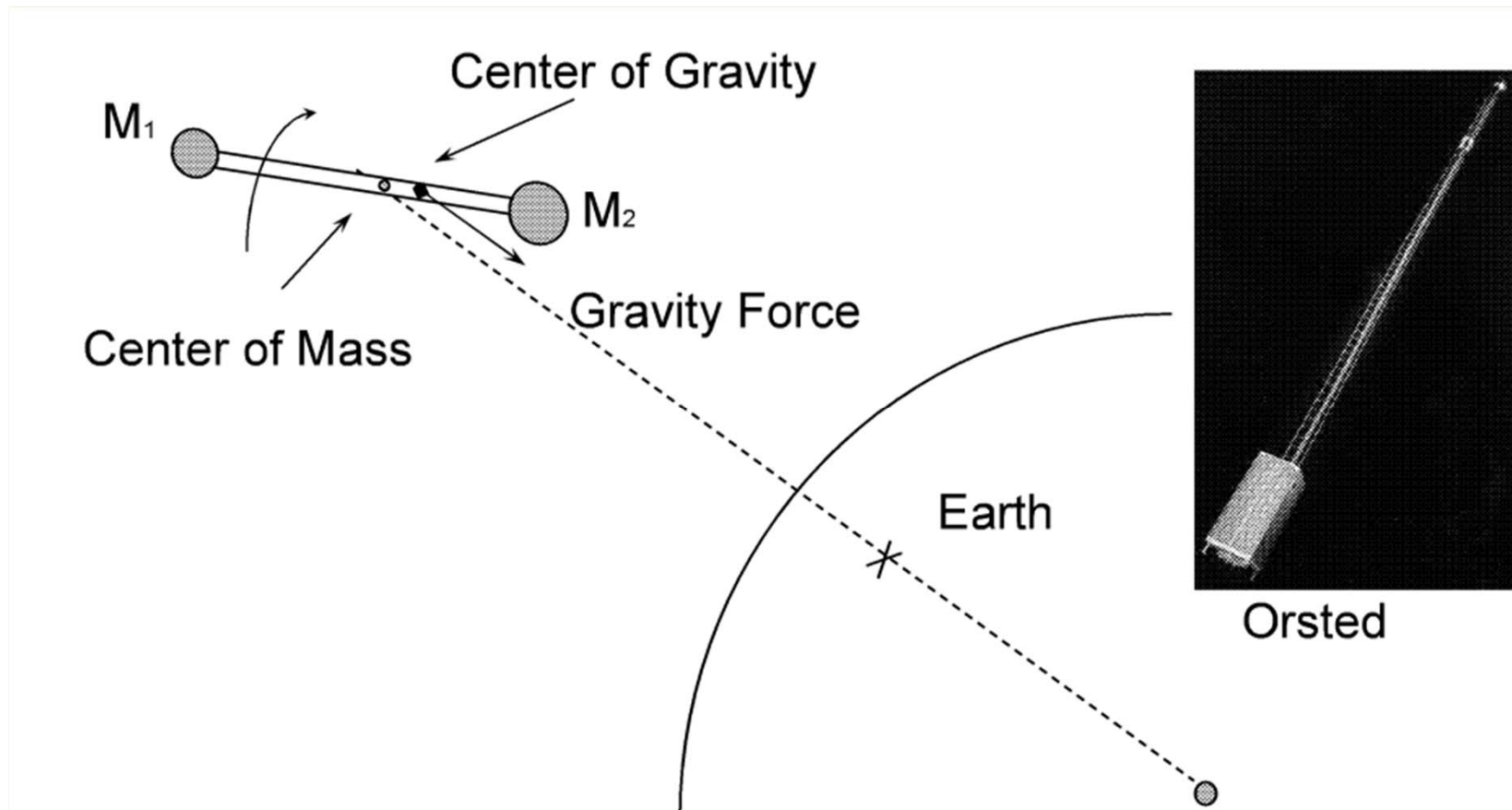
- Spin Stabilization
 - Gyroscopic Stiffness
- 3-axis Stabilization
 - Thruster
 - Momentum Exchange
 - Zero Bias/Bias Momentum



ST 1: Gravity Gradient

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- Only use *mass distribution* and *Earth gravity*
- Simple/Cheap



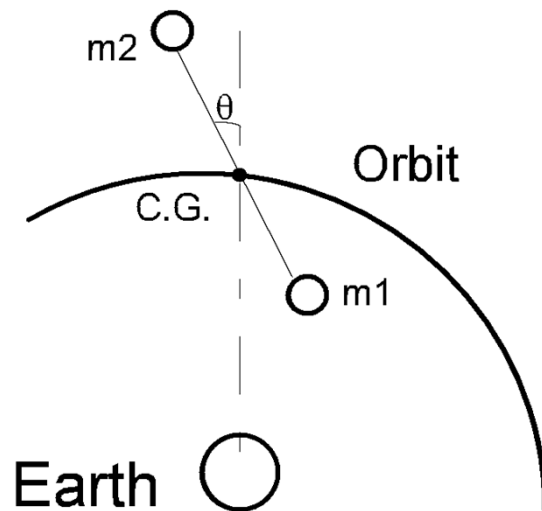
ST 2: Magnetic Torque

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- Use *Earth magnetic field*

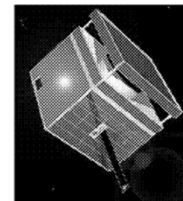
$$\vec{T} = \vec{M} \times \vec{B}$$

- M = Magnetic Dipole Moment
- B = Geomagnetic Flux Density

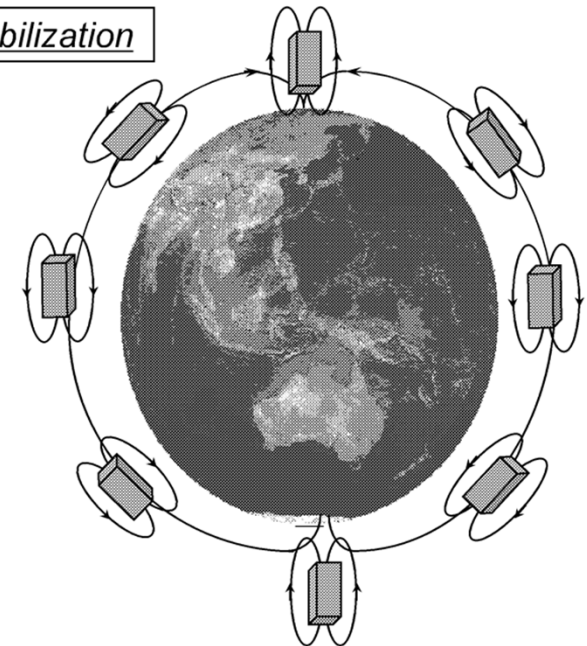


- Earth magnetic field is massively variable
- Only two axis control is available at any given time

Magnetic Stabilization



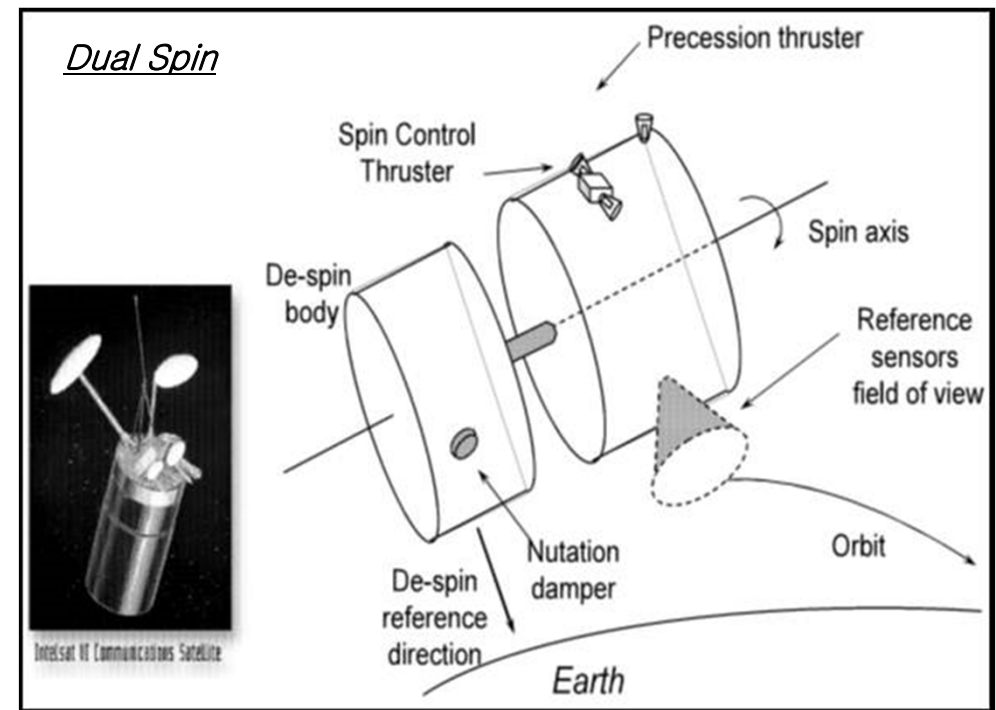
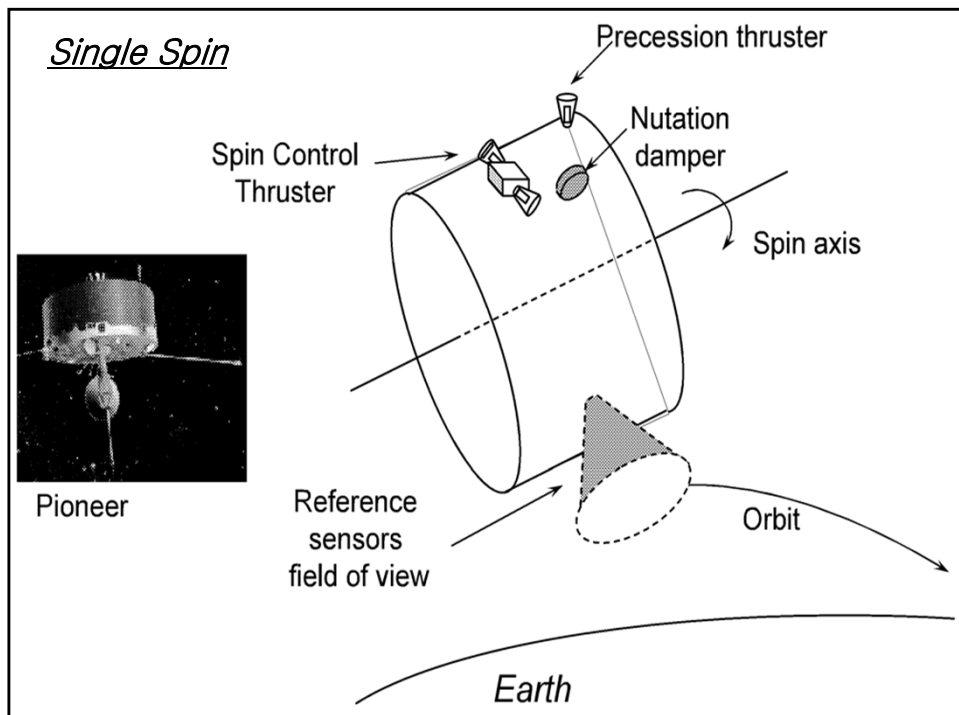
Munin



ST 3: Single/Dual Spin Stabilization

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- Spin to provide Gyroscopic Stiffness
 - Major/Minor axis $\leftrightarrow$ Minimum/Maximum Energy
 - Energy loss always drifts the spin axis toward the major axis
 - Not appropriate for extremely high precision pointing

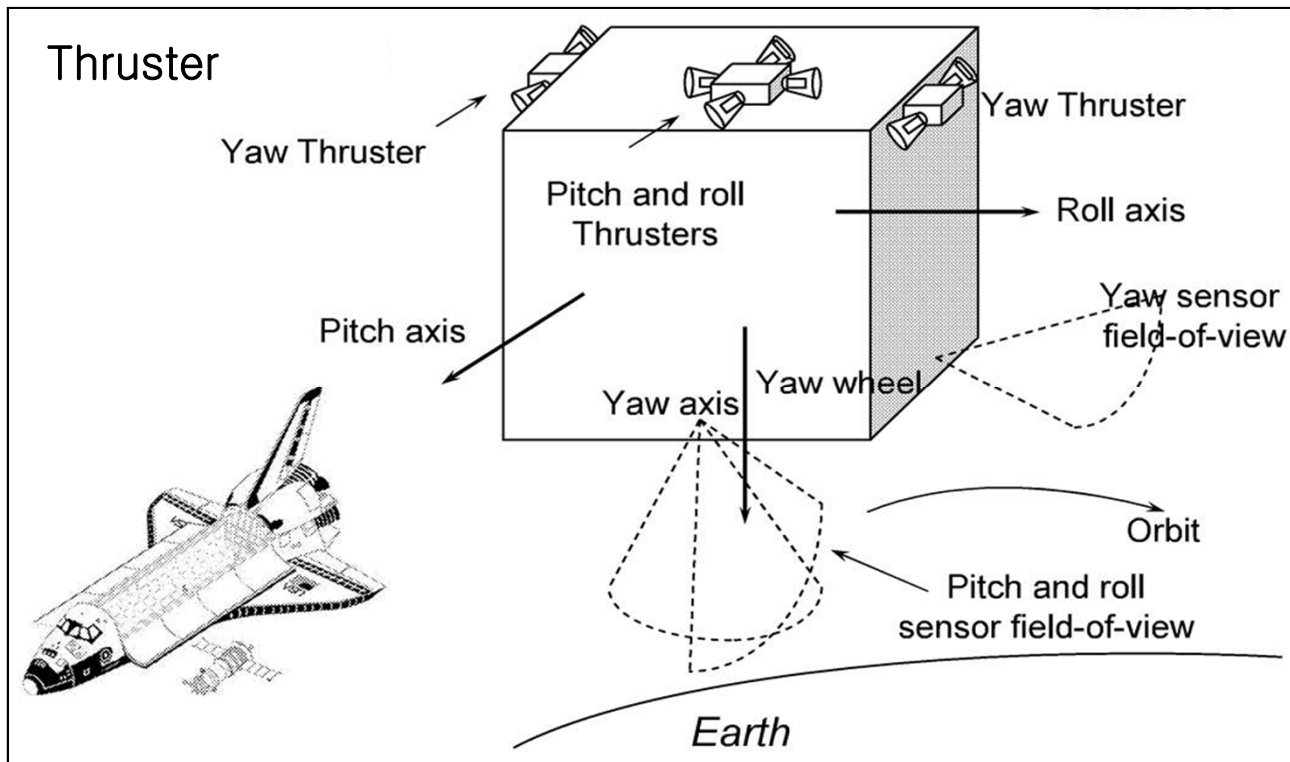


ST 4: 3-Axis Stabilization (Thruster)

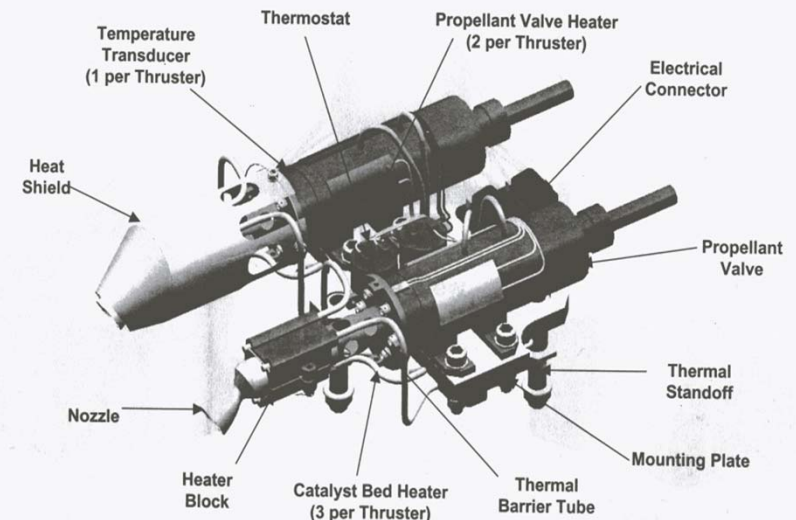
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$$\vec{T} = \vec{r} \times \vec{F}, \vec{F} = I_{sp} mg$$

- $\vec{r}$ = Moment Arm
- I_{sp} = Specific Impulse



• DTM (Dual Thruster Module)



ST 4: 3-Axis Stabilization – *Thruster*

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● Characteristics of Thrusters

Type	$I_{sp}(\text{sec})$	Thrust(lbf)	Comments
Cold gas(stored gas or liquid)	60	0.1~5	Typically N_2
Heated gas	100~800	0.1~5	Uses electric power or radioisotope
Liquid (Monopropellant)	220	1~100	Hydrazine(N_2H_4)
Liquid (Bipropellant) Storable/Cryogenic	280~310/ 350~440	5~10000	MMH/ N_2O_4 , H_2/O_2
Solid	280~300	~20000	
Ion (Solar electric)	1000~3000	m lbf	Use electric power
Solar Pressure (Sails)	infinite	μ lbf	

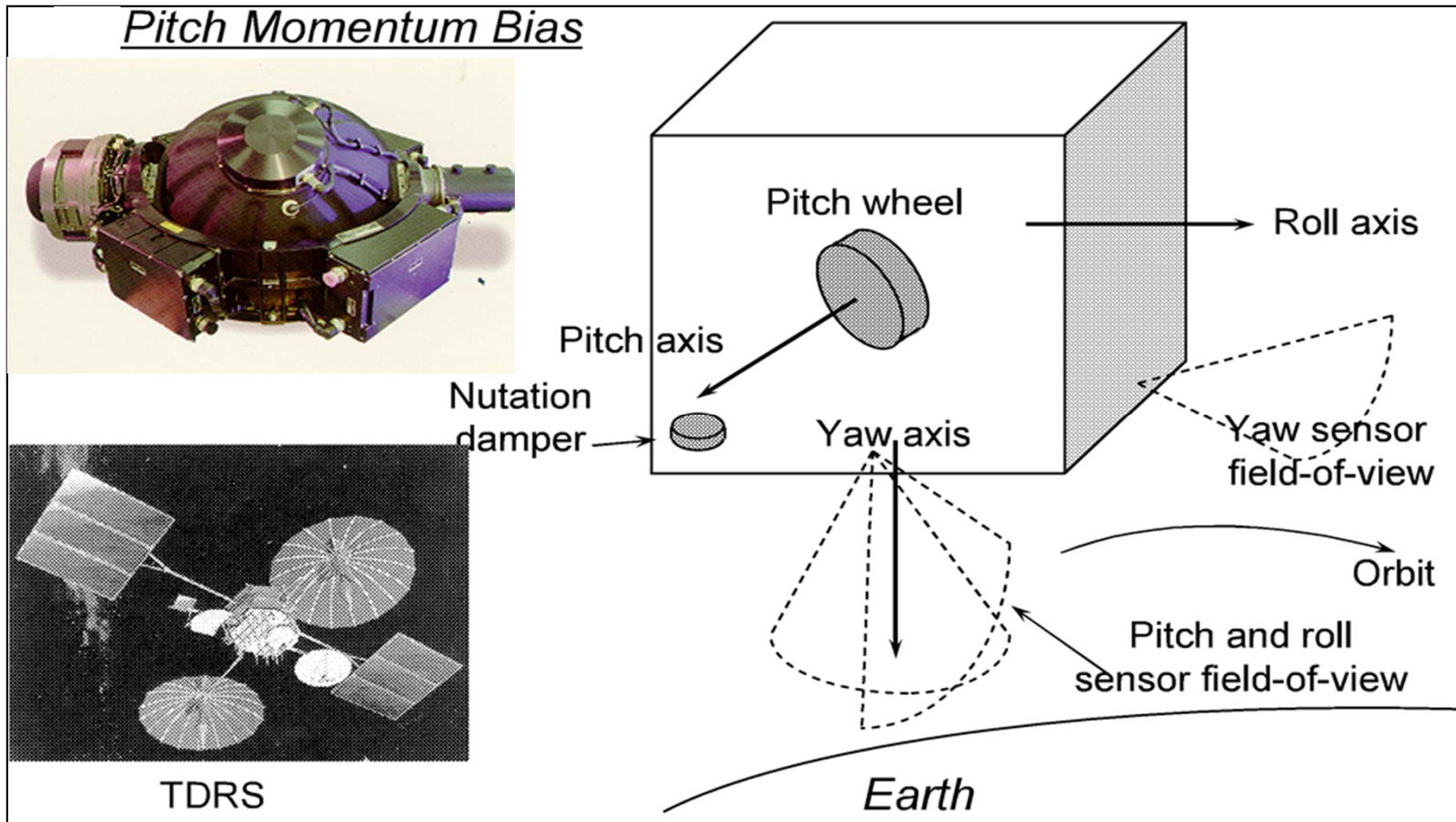
ST 5: 3-Axis Stabilization – *Momentum Exchange*

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- Biased Momentum Wheel

- Uni-directional Spin

Pitch Momentum Bias

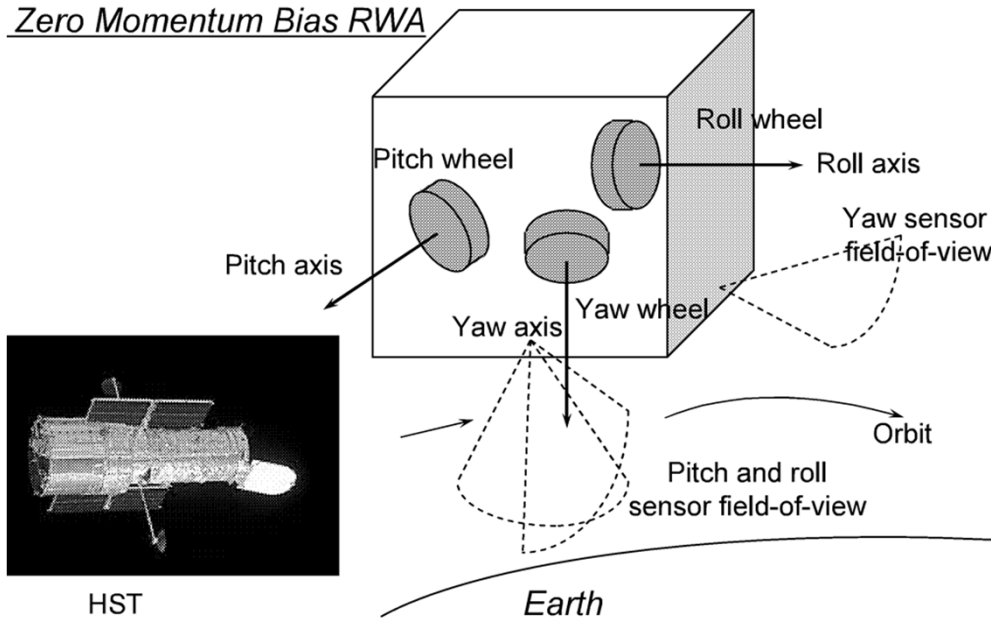


ST 5: 3-Axis Stabilization – *Momentum Exchange*

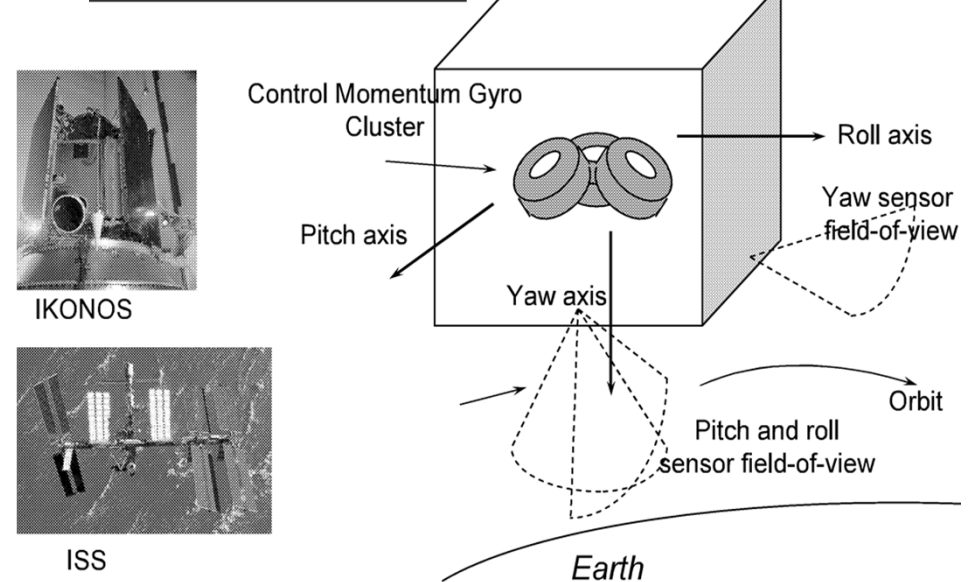
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- Zero-biased Reaction Wheel Assembly (RWA)
 - Bi-directional Spin
- Zero-biased Control Moment Gyro (CMG)
 - Wheel attached to steerable gimbal

Zero Momentum Bias RWA



Zero Momentum Bias CMG



Stabilization Techniques

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Control Method / Actuator	Accuracy (deg)	Torque Range (N-m)	Propellant Usage	Notes
Gravity Gradient	1 ~ 5	$10^{-6} \sim 10^{-3}$	N/A	passive, simple, cheap, central body oriented
Magnetic Torque	1 ~ 2	$10^{-2} \sim 10^{-1}$	N/A	cheap, slow, lightweight LEO
Single/Dual Spin Stabilization	0.1 ~ 1		High	passive, simple, cheap, internally oriented
Thruster	0.01 ~ 1	$10^{-2} \sim 10$		expensive, on/off control, quick response, consumable
Momentum Wheel	0.1 ~ 1		Low	expensive, similar to dual spin, spin in only one direction
Reaction Wheel	0.001 ~ 1	$10^{-1} \sim 1$	Low	expensive, precise faster slew, spin in either direction
Control Moment Gyro	0.001 ~ 1	$10^{-2} \sim 10^3$	Low	expensive, heavy, quick for fast slew, 3-axes

Reorientation Techniques

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● Linear Control

- Small angle maneuver
- PID Controller
- Linear Quadratic Controller



● Nonlinear Control

- Large Angle Maneuver
- Nonlinear effects of dynamics & actuators
- Feedback Linearization
- Robust Control
 - H-infinity control
 - Sliding mode control
- Adaptive Control
- Soft Control
 - Neural Network
 - Fuzzy Logic
- Optimal Control

RT 1: Euler Angle Feedback (Small Angle)

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- Linearized Equations of Motion

- Small Euler Angle rotations
- Principal Axis

$$\begin{aligned} J_1 \ddot{\phi} &= T_{cx} + T_{dx} \\ J_2 \ddot{\theta} &= T_{cy} + T_{dy} \\ J_3 \ddot{\psi} &= T_{cz} + T_{dz} \end{aligned}$$



- Simple PD control law

$$\begin{aligned} T_{cx} &= K_x (\phi_{com} - \phi) + K_{xd} \dot{\phi} = K_x \phi_E + K_{xd} \dot{\phi}, \\ T_{cy} &= K_y (\theta_{com} - \theta) + K_{yd} \dot{\theta} = K_y \theta_E + K_{yd} \dot{\theta}, \\ T_{cz} &= K_z (\psi_{com} - \psi) + K_{zd} \dot{\psi} = K_z \psi_E + K_{zd} \dot{\psi} \end{aligned}$$

RT 2: Quaternion Feedback (Large Angle)

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- Well-suited for onboard real-time implementation
- Kinematic/Dynamic Equations of Motion

$$\begin{array}{l} \dot{\mathbf{q}} = -\frac{1}{2}\boldsymbol{\omega} \times \mathbf{q} + \frac{1}{2}q_4\boldsymbol{\omega}, \quad \dot{q}_4 = -\frac{1}{2}\boldsymbol{\omega}^T \mathbf{q} \\ \hline J \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J \boldsymbol{\omega} = \mathbf{u} \end{array} \quad \mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}, \quad \boldsymbol{\omega} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

- Globally Asymptotically Stable Controller:

- Linear State Feedback

$$\mathbf{u} = -K\mathbf{q} - C\boldsymbol{\omega}$$

$$K = [\alpha J + \beta I]^{-1}, \quad \alpha, \beta \geq 0, \quad K^{-1}C > 0$$

- Counteraction of Gyroscopic Term:

$$\mathbf{u} = -K\mathbf{q} - C\boldsymbol{\omega} + \boldsymbol{\omega} \times J\boldsymbol{\omega}, \quad K^{-1}C > 0$$

- PID controller applied to many S/C missions

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